

1 Introduction

Large language models (LLMs) provide a promising interface for task decomposition, contextual reasoning, and long-horizon planning in robotics. With structured prompts, constrained action interfaces, or formal specifications, LLMs can produce increasingly executable robot plans [1, 2]. Indoor mobile robots, however, must continually balance task progress against safety risks arising from obstacles, narrow passages, uncertain observations, and resource limitations. An LLM may overlook rare but severe failures in the execution history or make inconsistent recommendations in similar situations. More fundamentally, a semantically plausible explanation does not imply that an action satisfies physical and operational constraints. Exploiting long-term experience for risk assessment without allowing the accumulated history to overwhelm the model context is therefore a central challenge for deploying language agents in real robot loops.

Prior language agents learn from interaction histories through non-parametric memory. Reflexion stores verbal reflections derived from task feedback, ExpeL extracts transferable natural-language insights from collections of experiences, and Voyager maintains a reusable library of embodied skills [3, 4, 5]. These studies demonstrate that experience can improve subsequent behavior without updating model parameters, but they primarily target general task performance. Retaining complete robot trajectories in the prompt causes the input to grow with deployment time, whereas storing only highly abstract summaries can remove the evidence from which a lesson was derived. Moreover, conventional retrieval typically returns one generic set of similar cases to a single decision maker. It does not account for the asymmetric evidence needs of deliberative roles: a task advocate benefits from successful routes and effective mitigations, while a risk critic should prioritize failures, high-severity hazards, and stopping conditions.

Complementary work constrains unsafe LLM plans using temporal logic, multi-model safety feedback, or formal guardrails. SELP enforces temporal specifications during plan generation, SAFER couples a task planner with a safety agent, and RoboGuard contextualizes safety rules before resolving conflicts through control synthesis [2, 6, 7]. These approaches substantially advance safe LLM-based robot planning, but historical risk knowledge is not their primary focus. In particular, existing systems do not jointly address three related requirements: compressing long-term risk history at multiple abstraction levels, retaining a verifiable evidence chain from reusable rules to source episodes, and providing role-specific rather than homogeneous historical evidence during deliberation.

We present a risk-experience-aware multi-role agent framework for indoor mobile robots. It organizes long-term interaction experience through hierarchical risk memory, limiting context growth while preserving the connection between abstract safety knowledge and source cases and allowing the empirical safety boundary to evolve with new evidence. Role-conditioned retrieval then supplies differentiated historical evidence to a Task Advocate, a Risk Critic, and a Safety Decision Agent, enabling complementary deliberation over task utility and safety risk. The agents produce only structured semantic recommendations; deterministic constrained optimization and an independent Safety Guard retain final execution authority,

confining language reasoning to an auditable safety boundary. The main contributions of this work are:

1. A risk-experience-aware multi-role agent architecture that integrates historical evidence through complementary Task Advocate, Risk Critic, and Safety Decision roles, while separating language reasoning from execution authority.
2. A hierarchical safety-risk memory based on Risk Memory Cards that jointly represents environmental state, task objective, action outcome, failure cause, and risk factors, with L0–L1–L2 abstraction for compression, traceability, and safety-boundary evolution.
3. A systematic evaluation protocol spanning multi-scenario closed-loop simulation, component ablations, and real-robot experiments, targeting task performance, risk response, decision efficiency, and long-term safety-boundary updates.

Our scope is high-level semantic action selection rather than continuous trajectory-level safety guarantees. The reported risk and uncertainty scores support interpretable relative comparisons and are not treated as calibrated collision probabilities.

References

- [1] I. Singh et al., “ProgPrompt: Generating Situated Robot Task Plans using Large Language Models,” *ICRA*, 2023.
- [2] Y. Wu et al., “SELP: Generating Safe and Efficient Task Plans for Robot Agents with Large Language Models,” *ICRA*, 2025.
- [3] N. Shinn et al., “Reflexion: Language Agents with Verbal Reinforcement Learning,” *NeurIPS*, 2023.
- [4] A. Zhao et al., “ExpeL: LLM Agents Are Experiential Learners,” *AAAI*, 2024.
- [5] G. Wang et al., “Voyager: An Open-Ended Embodied Agent with Large Language Models,” arXiv:2305.16291, 2023.
- [6] A. A. Khan et al., “Safety Aware Task Planning via Large Language Models in Robotics,” *IROS*, 2025.
- [7] Z. Ravichandran et al., “Safety Guardrails for LLM-Enabled Robots,” *IEEE RA-L*, 2026.